Radiation and the Atmosphere

Cosmic Backgrounds

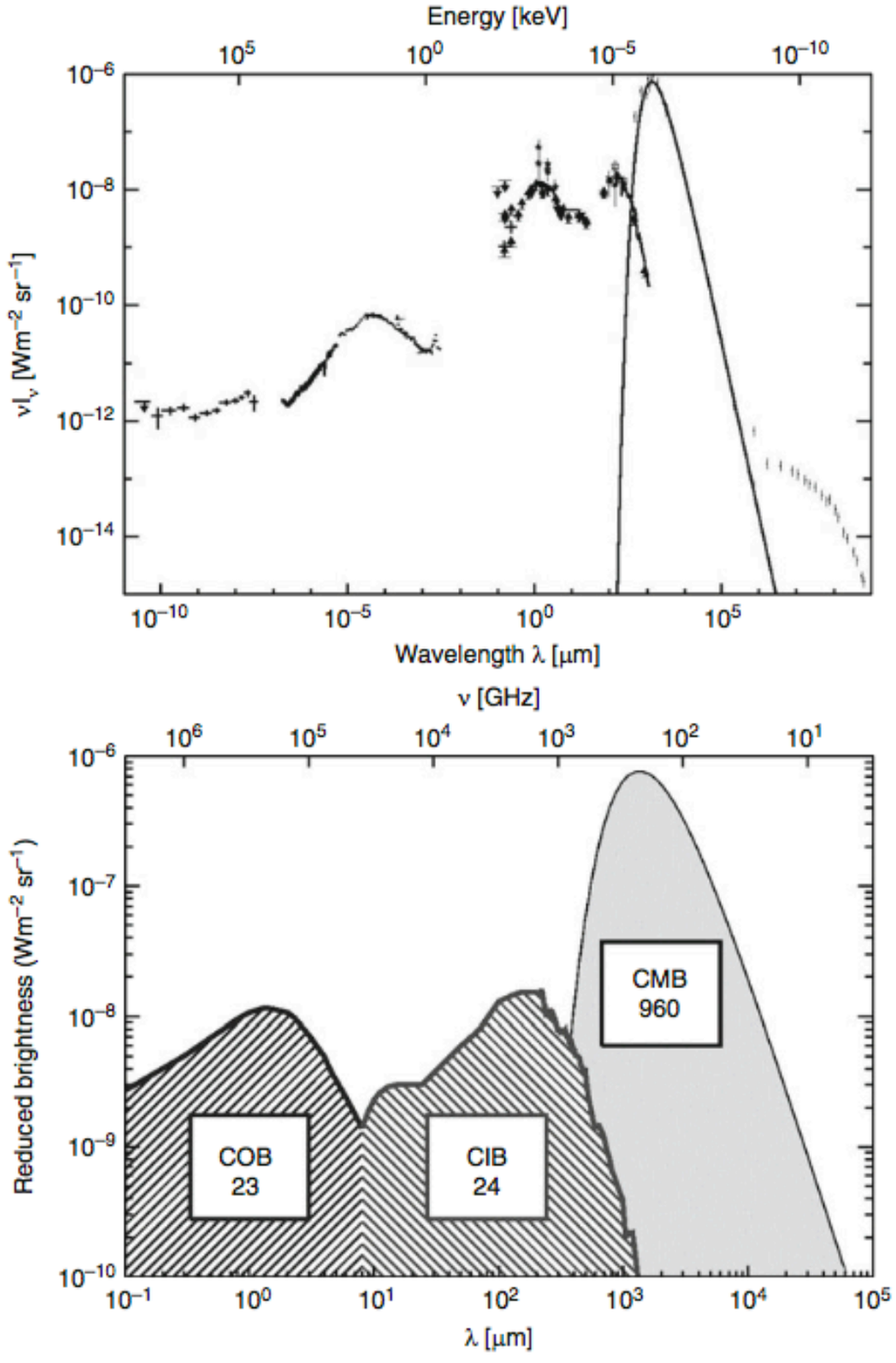


Fig. 3.2 *Upper*: Observed sky background radiation, plotting reduced brightness νI_ν against wavelength (*lower scale*) or against photon energy (*upper scale*). From left to right are the extragalactic ultraviolet, visible, and infrared contributions, the cosmological blackbody contribution, and finally the radiofrequency radiation in the Galaxy. The interplanetary component (zodiacal emission) has been subtracted. *Lower*: Smoothed extract from the above spectrum, showing the cosmic optical background (COB), cosmic infrared background (CIB), and cosmic microwave background (CMB). Numerical values indicate the total brightness of each component in $nW\,m^{-2}\,sr^{-1}$. Note that the infrared background (formation and evolution of galaxies) is comparable with the optical background (stars), and that the latter is a small fraction of the microwave background. (Figures with the courtesy of Hervé Dole, private communication, 2007)

Radiation: Definitions

- Imagine there is a source in the sky emitting monochromatic intensity as a function of frequency, v:

$$I_\nu(\alpha, \delta)$$

- Units: W/m^2/sr/Hz

- Or, in photons/s/m^2/sr/Hz,

$$N_\nu(\theta) = \frac{I_\nu(\theta)}{h\nu}$$

Radiation: Definitions

- The monochromatic flux (a function of frequency, ν) received by a surface (e.g. a telescope mirror) is:

$$\mathscr{E}_\nu = \int I_\nu \cos(\theta) d\Omega$$

- where θ is the angle between the normal to the surface and the direction of the ray and Ω is the solid angle subtended by the source

- (For a telescope pointing at an astronomical object, cos θ ~ 1)

- This is the notation in the textbook, but I will often use *f* for flux

- Units: $W/m^2/Hz$

Radiation: Definitions

- Both of these quantities can also be expressed per unit wavelength:

$$I_\lambda \qquad \mathscr{E}_\lambda$$

- SI Units: W/m²/**sr**/nm and W/m²/nm

- Since

$$I_\nu d\nu = I_\lambda d\lambda$$

- and $c = \nu\lambda$

$$I_\lambda = I_\nu \nu^2 / c$$

Radiation:Example

$$B_\nu(T) = \frac{2\nu^2 k_\mathrm{B} T}{c^2}$$

Blackbody in Rayleigh-Jeans limit (long wavelength or low frequency)

$$B_\lambda = B_\nu \nu^2 / c$$

$$B_\lambda(T) = \frac{2ck_\mathrm{B}T}{\lambda^4}$$

But we do not measure this radiation perfectly

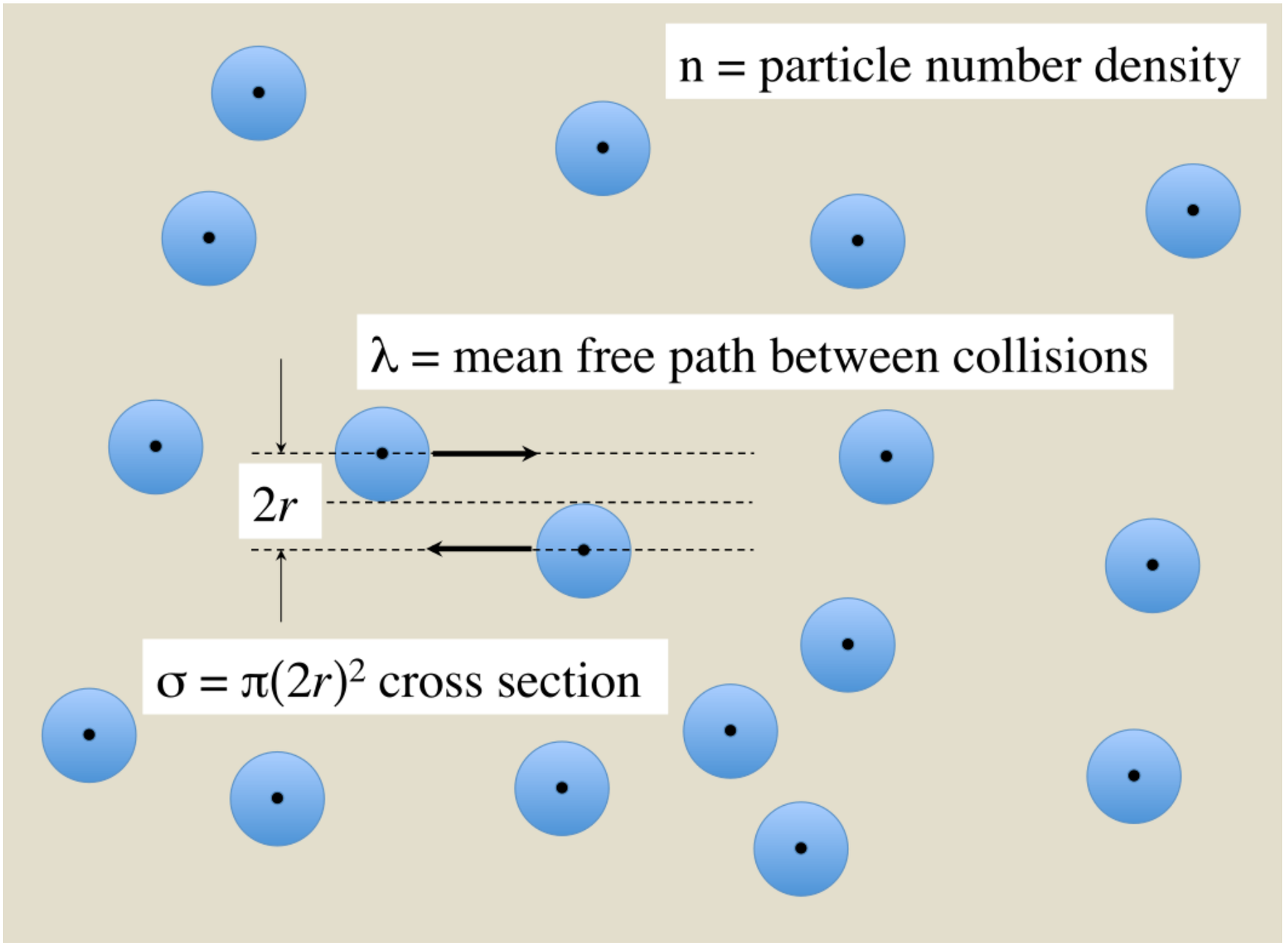
- Aberrations: imperfections in optical surfaces
- Diffraction: finite size of telescopes, antennas
- Noise: finite number of photons (inc. contribution from sky, readout etc).
- Finite spectral resolution of spectrographs

Foregrounds and backgrounds

- Atmosphere
  - Blurring (“seeing”): *forward model, AO*
  - Extinction (absorption and scattering): *can calibrate*
  - Emission (airglow, thermal in IR): subtract
- Astronomical
  - Dust extinction: *can calibrate*
  - e.g. radio emission contaminating CMB: *can subtract?*

Why is the sky blue?

- … same reason the Sun is red.
- Extinction is absorption and scattering: both remove light from a bright source
- In the case of the optical sky, it is mostly *Rayleigh scattering*, which is proportional to $1/\lambda^4$.
- Of course this also affects stars and galaxies! They are dimmer and redder at close to the horizon.



By Qwerty123uiop, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=62405377

Extinction

- Consider a thin slice of this with thickness *dl* and area *A*.
- In a slice: $N_{\text{particles}} = nAdl$
- probability of collision $p = \dfrac{\sigma N}{A} = \sigma n dl$
- At each layer, the flux is reduced by a factor $\dfrac{df}{f} = -p = -\sigma n dl$
- Integrate both sides

$$\int_{f_0}^{f} \frac{df'}{f'} = -\int \sigma n(l) dl$$

$$\ln(f/f_0) = -\sigma \int n(l) dl \equiv -\tau$$

$$f/f_0 = e^{-\tau}$$

- where we have defined the optical depth τ

Airmass

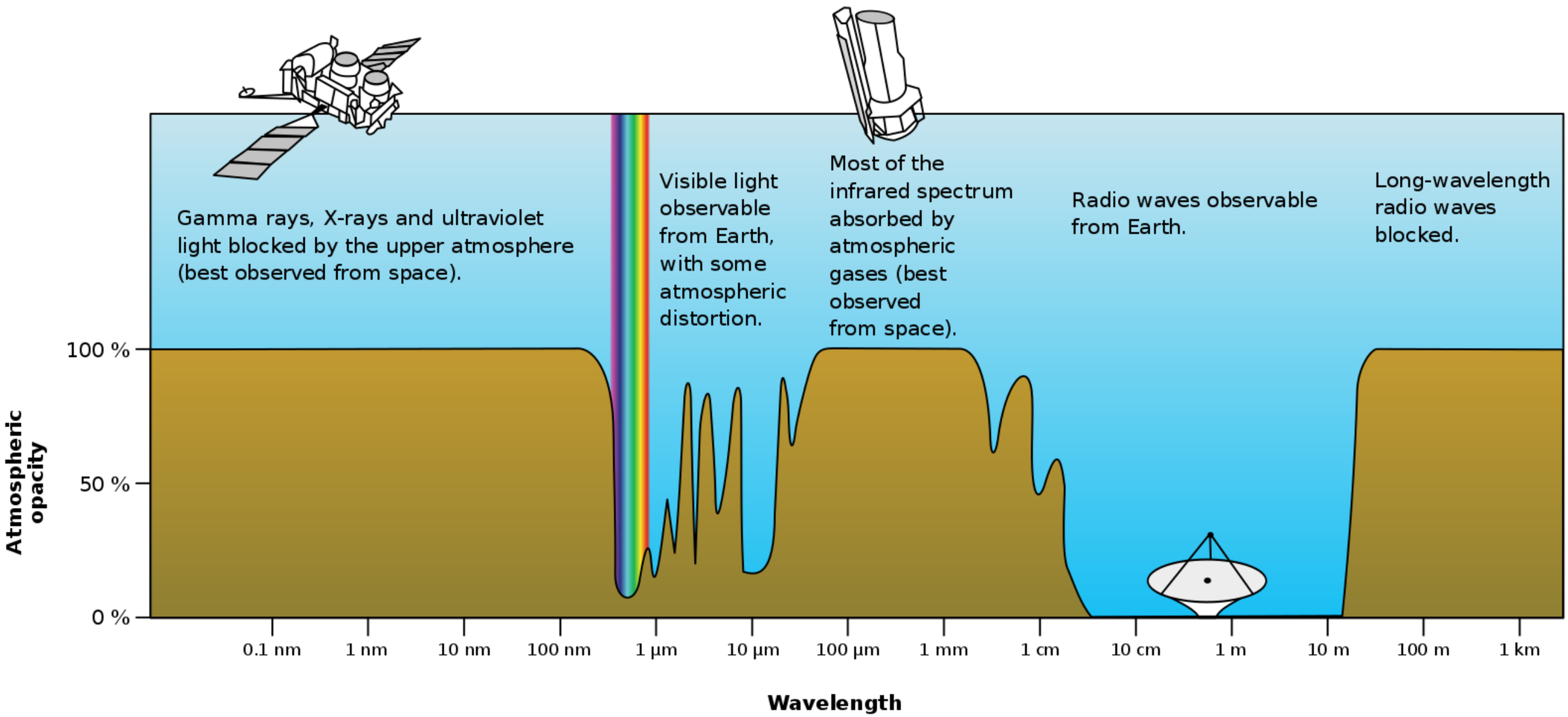
- To a good approximation, the path length through the atmosphere can be calculated by treating the atmosphere like a flat slab.

- The path length scales as the secant of the angle from the zenith (directly overhead).

- (This breaks down at large zenith angles, but one rarely wants to observe there)

- By convention, Airmass = 1 at the zenith.

Atmospheric Windows



Atmospheric absorption

- In other wavelengths, much of the sky is opaque due to absorption by molecules.

- This is especially true in the IR due to e.g. CO_2, H_2O, OH etc.

Atmospheric windows in IR and submm

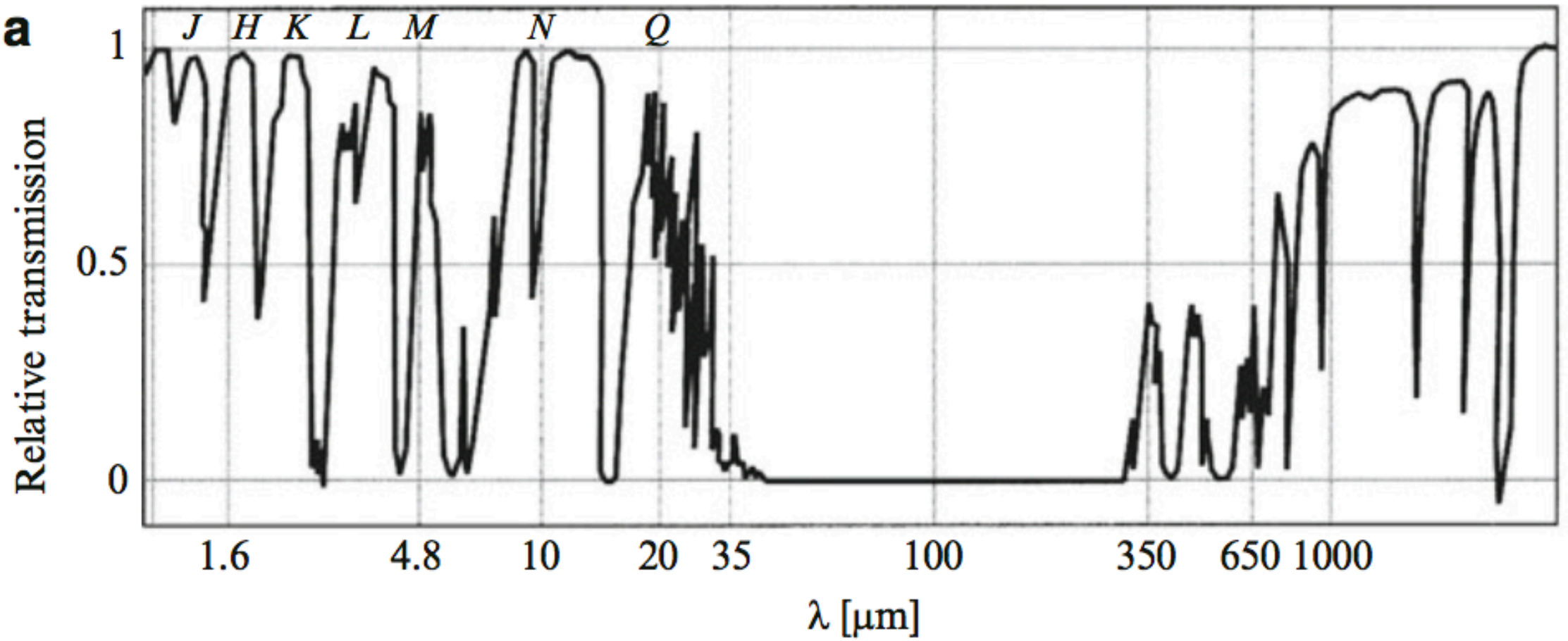


Fig 2.8 Atmospheric **transmission** (at altitude 4,200 m) across the whole infrared and submillimetre range. Spectral bands are indicated with the usual notation. The relatively good submillimetre transparency corresponds to a quantity of precipitable water less than 1 mm.

Atmospheric “Back”grounds I: Sky in the Optical

- The same extinction effect that makes the sky blue by day, also makes the night sky not completely black at the optical wavelengths

- The effect is worse sat short wavelengths (ultraviolet) than at long (near infrared)

- Data from Cerro Tololo (Chile)

Table I
Sky Brightness as a Function of Lunar Phase

Days from	Sky Brightness (mag/arcsec2)				
New Moon	U	B	V	R	I
0	22.0	22.7	21.8	20.9	19.9
3	21.5	22.4	21.7	20.8	19.9
7	19.9	21.6	21.4	20.6	19.7
10	18.5	20.7	20.7	20.3	19.5
14	17.0	19.5	20.0	19.9	19.2

Atmospheric Backgrounds II: Thermal

- The Earth, the telescope and the atmosphere are ~blackbodies with T ~ 270 K.

- Emission from these creates a background which rises longer than a few microns and peaks in the range 10-20 microns.

- (The detectors themselves are cryogenically cooled)

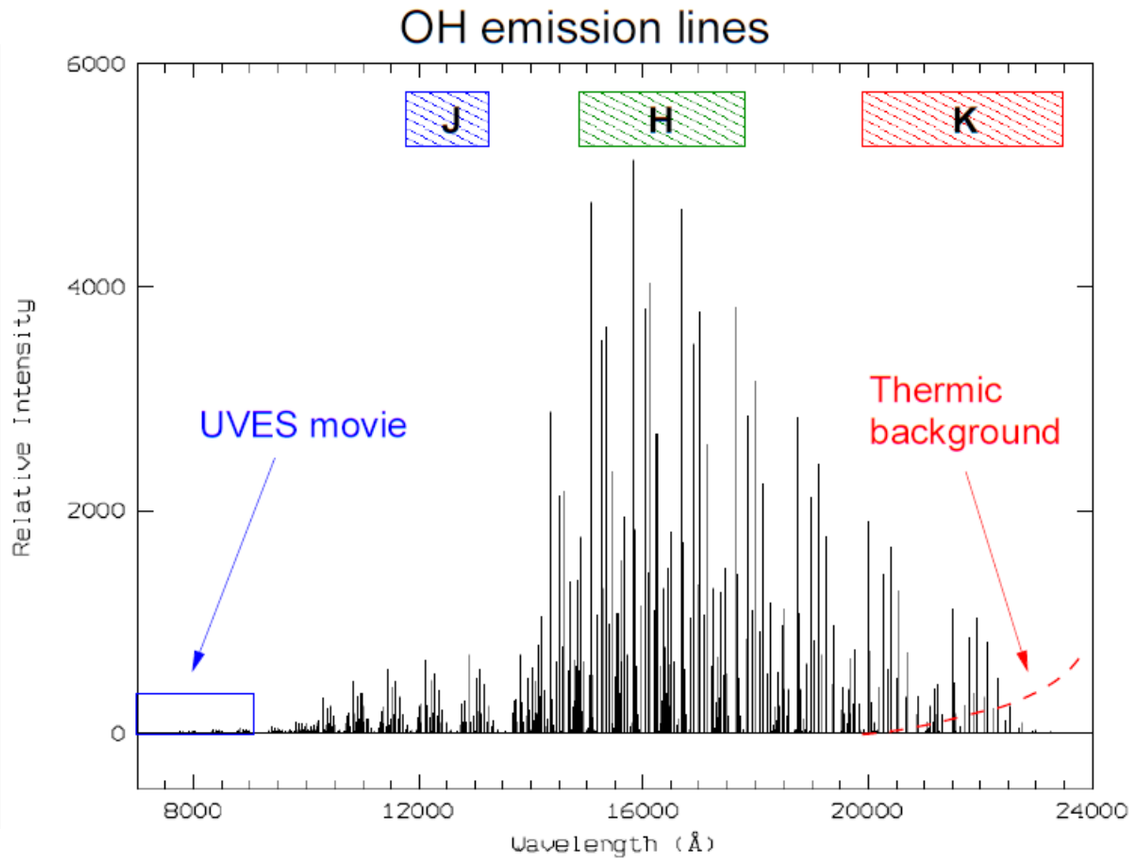
Atmospheric Backgrounds III: Airglow



Credit: P. Horálek/ESO

Atmospheric Backgrounds III: Airglow





- Various excitations are driven directly or indirectly by solar radiation. Similar to aurorae

- Movie credit: Ferdinando Patat (ESO)

Dealing with variable night sky: mid-IR through submm

- Because the sky brightness is highly variable in the mid-IR and longer wavelengths, one strategy for imaging is to “chop” the secondary mirror on a rapid timescale (Hz): this moves the field to an off-target area of blank sky.

- One can then subtract an image of the sky taken at (almost) the same time as the exposure

- But as the secondary is chopped it also sees the emission from the telescope differently.

  - Solution: nod the telescope

- This leads to complicated scanning strategies

Planning Observing

- Optical:
    - Observations of faint galaxies are limited by the brightness of the sky
    - Do you need dark time or is grey time OK?
    - *u* band or *i* band?
    - Especially if *u*, also consider zenith angles

Atmospheric Refraction

- Because the index of refraction is not 1, the atmosphere acts like prism

- Fixed with an Atmospheric Dispersion Corrector (ADC), essentially prisms which reverse the dispersion from the atmosphere

Atmospheric Turbulence

- Turbulence causes spatial and temporal variations in the atmosphere
- Astronomical images are blurred out.
- This can be characterized by a “point spread function” (PSF) which can be estimated from bright stars.
- Full-width-half-max (FWHM), i.e the angular diameter at half height of the star image profile is used to characterize the “seeing.”
  - 0.5” excellent seeing, rarely gets better than this
  - 1.0” typical seeing at a good site
  - 3.0”+ Waterloo
    - *compare to JWST 0.05”*
- More on Adaptive Optics later